



IEA Bioenergy

Technology Collaboration Programme



Gasification developments in Europe and the USA

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(BOKU)

IEA Bioenergy Task 33 webinar

24. February 2021

The IEA Bioenergy Technology Collaboration Programme (TCP) is organised under the auspices of the International Energy Agency (IEA) but is functionally and legally autonomous. Views, findings and publications of the IEA Bioenergy TCP do not necessarily represent the views or policies of the IEA Secretariat or its individual member countries.

Technology Collaboration Programme

by **iea**

Outline

- IEA Bioenergy Task 33
- Status on gasification in
 - Austria
 - Germany
 - Italy
 - NL
 - Sweden
 - UK
 - USA
 - France
 - Spain



Figure source: SynCraft

IEA Bioenergy Task 33 - Gasification of biomass and waste

www.task33.ieabioenergy.com

Task33

Gasification of Biomass and Waste

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IEA Bioenergy

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[Gasification explained](#) [Task 33 Description](#) [Participants and Country Reports](#) [Task Meetings and CR updates](#) [Workshops and Events](#) [Task 33 Projects](#) [Other Publications and Reports](#) [Links](#) [Videos](#)

Welcome

Task 33 is a working group of international experts with the aim to promote the commercialization of efficient, economical and environmentally preferable thermal biomass gasification processes.

Latest Updates

2021-02-11 | Events
Task 33 Webinar: "Gasification: A Crucial Technology for the Energy Transition. A Global Perspective"

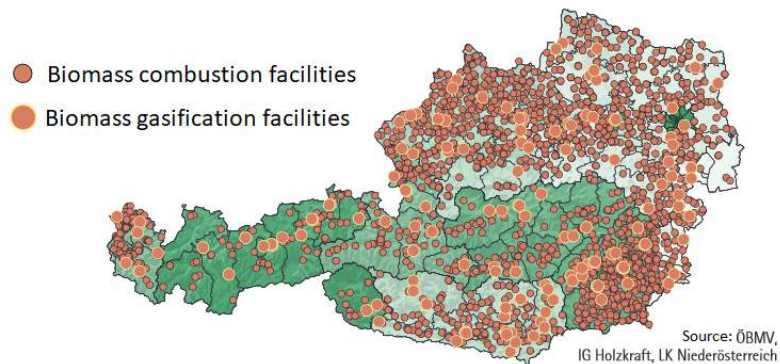
24. February 2021

11.00 am - 01.00 pm (CET)

›› Read more

DISCLAIMER:
The Task 33 / Thermal Gasification of Biomass, also known as the Task 33 / Thermal Gasification of Biomass, functions within a framework created by the International Energy Agency (IEA). Views, findings and publications of Task 33 / Thermal Gasification of Biomass do not necessarily represent the views or policies of the IEA Secretariat or of all its individual member countries.

Austria



About 2 400 biomass boilers and more than 140 CHP gasification facilities in Austria.

All facilities are small scale CHP.
Besides Austrian facilities also facilities by German and Italian manufacturers.



Output kW _{el}	Output kW _{th}
18/55	44/120
20	60
50	107



200-500	320-770
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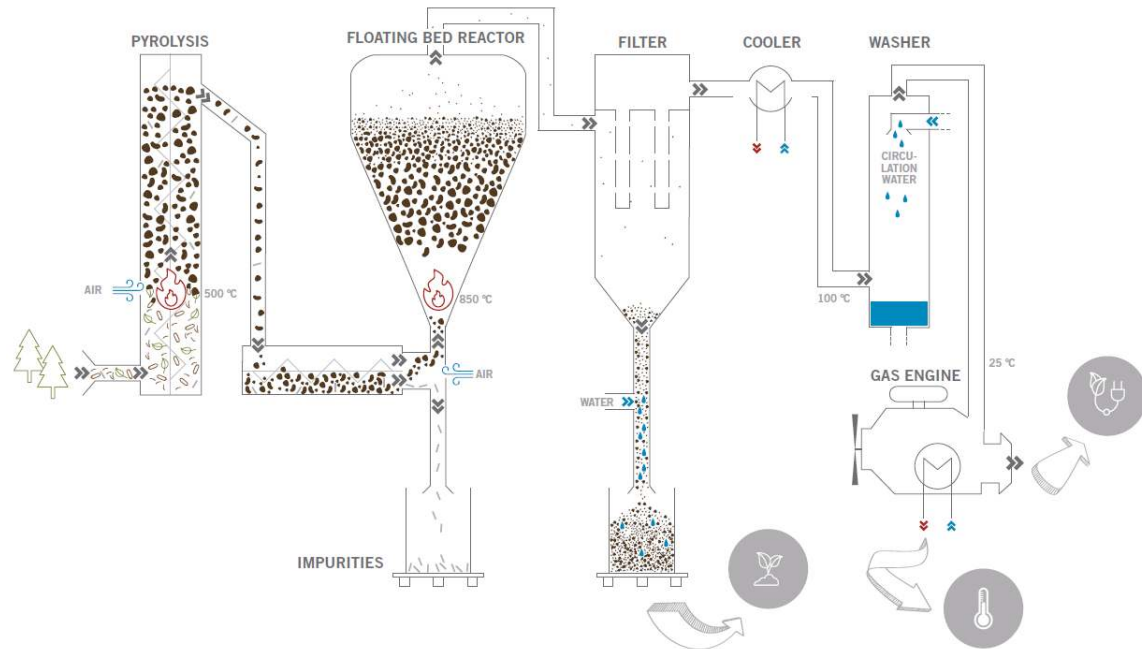


300/500	500/800
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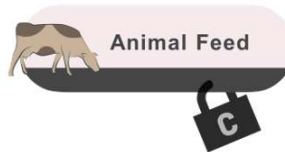
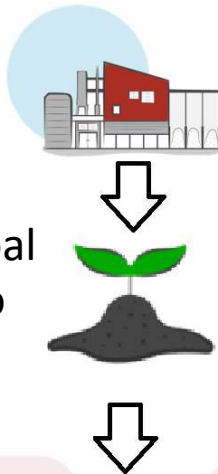


120-550	280-880
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Austria



1kg charcoal
is an eq. to
3 kg CO₂



Floating fixed bed technology



Mission

negative CO₂ footprint

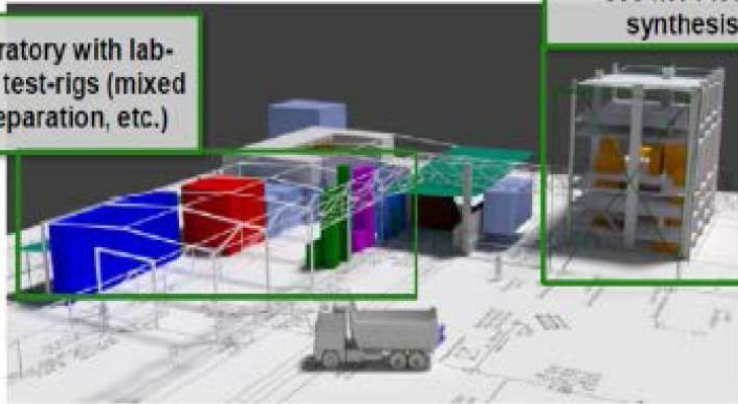
Austria: Project Waste2Value

1 MW DFB gasification + synthesis: Waste2Value

New research location at industrial site *Simmeringer Halde*

Research laboratory with lab-scale synthesis test-rigs (mixed alcohols, H₂ separation, etc.)

1 MW DFB gasification demonstration plant (improved reactor design) + 300 kW Fischer-Tropsch synthesis pilot plant



COMET

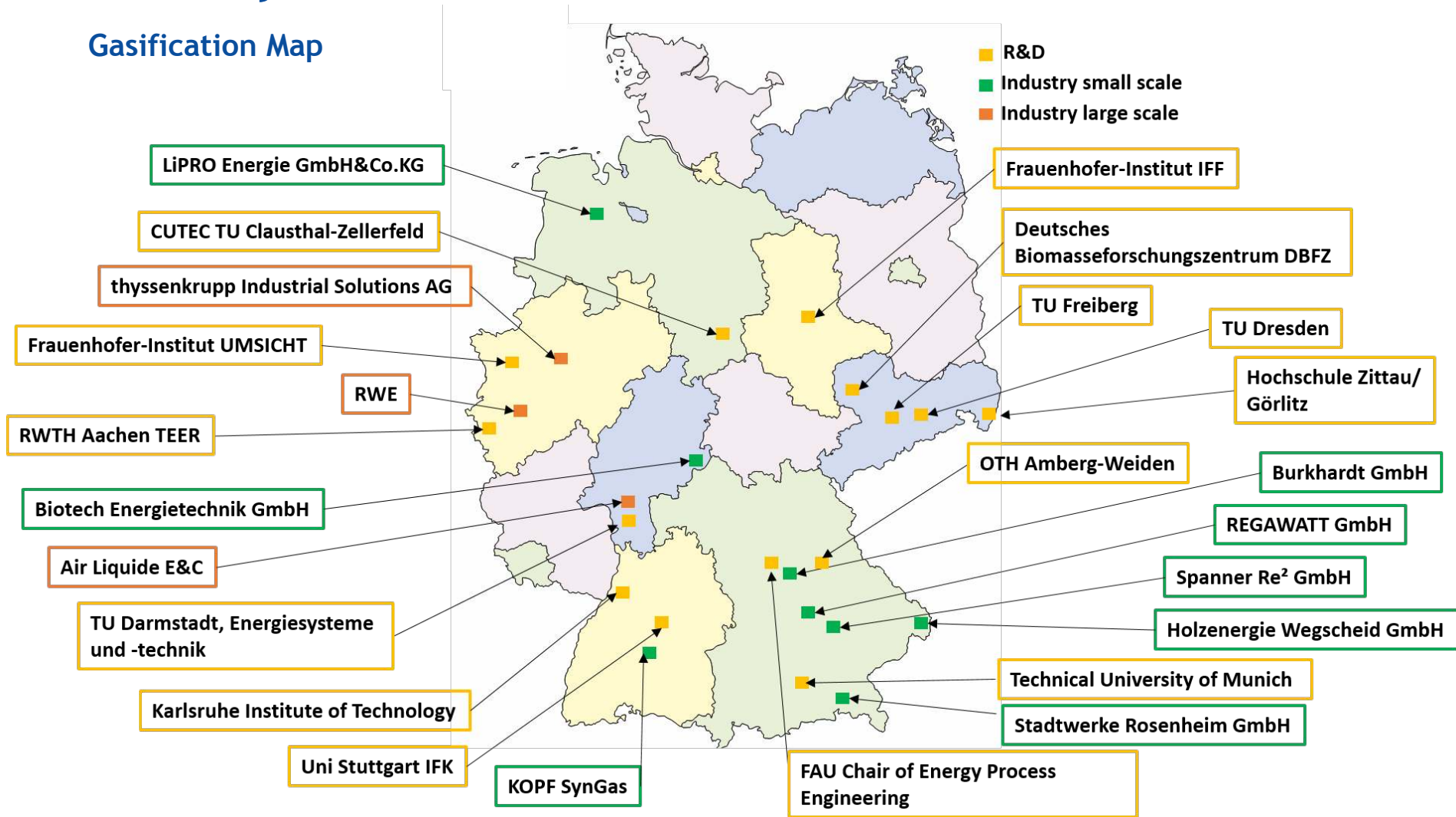
Competence Center for
Excellent Technologies

Now under construction
April 2021 first tests planned
May 2021 commissioning
June 2021 operation



Germany

Gasification Map



Germany

Bioliq pilot plant



Developed at KIT for production of synthetic biofuels and base chemicals from biogenic residues,

Process chain (TRL6) with:

2 MW fast pyrolysis for biosyncrude

5 MW entrained flow gasifier (40/80 bar) with high temperature gas cleaning

fuel synthesis: DME to gasoline (100 l/h)

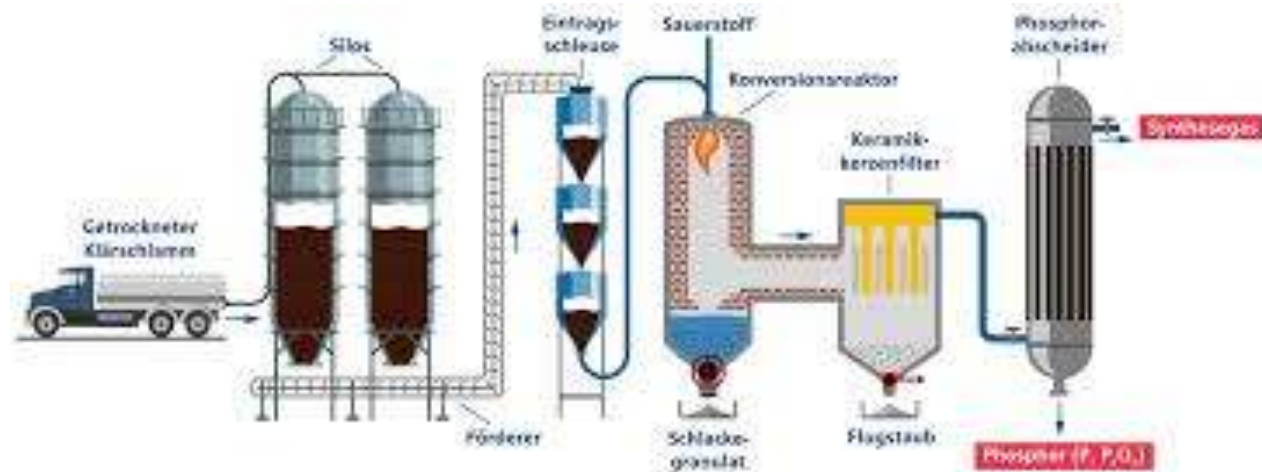
Thermo-chemical process steps in operation since 2010, synthesis since 2014

Operation 24/7 in campaigns

Germany

RWE Sewage Sludge to Phosphorus

Pilot plant in Niederaußem



Entrained flow gasifier

Atmospheric, refractory lined, dip quench, liquid ash discharge

Feedstock: Sewage Sludge, Sewage Sludge Ash, Lignite 130 kg/h

Temperature ~1500°C

Start-Up April 2021

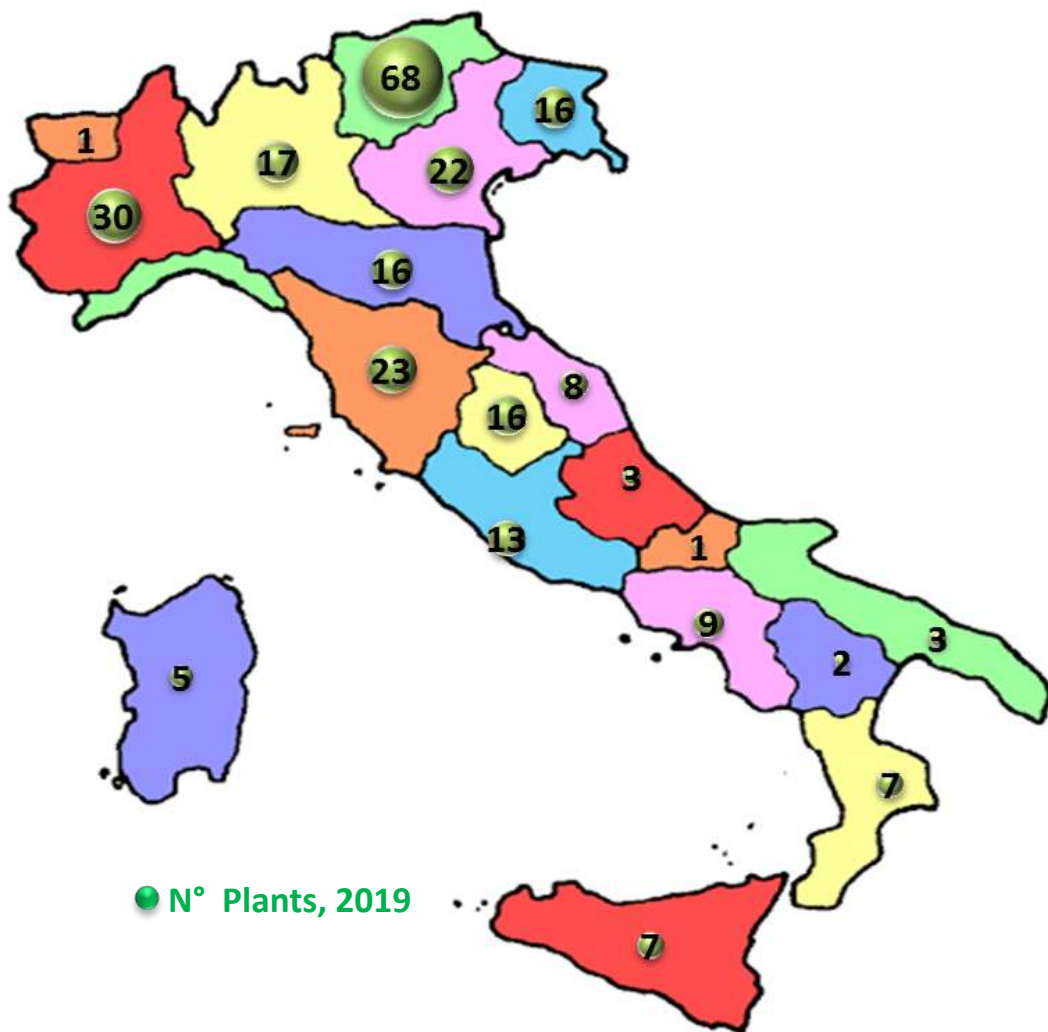
Funding provided by State of North Rhine Westphalia (Ministry of Economics);

Total project budget (incl. cost for plant operation): 6.7 Mio. €

- Partners: Fraunhofer UMSICHT, Ruhr Universität Bochum

Source: RWE; <https://www.group.rwe/presse/rwe-power/2019-08-01-neueversuchsanlage-gewinnt-lebenswichtigen-rohstoff-phosphor-aus-klaerschlamm-zurueck>

Italy - Gasification plants across Italy - 2019

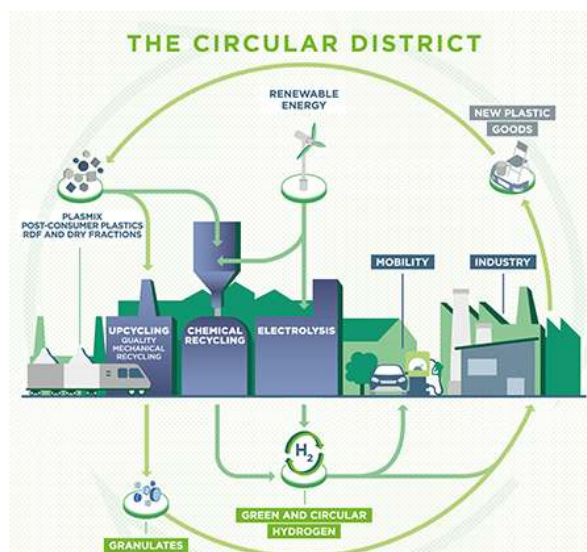


Year of ref.	2019
Total Power (kWel)	56674
Total Number	267

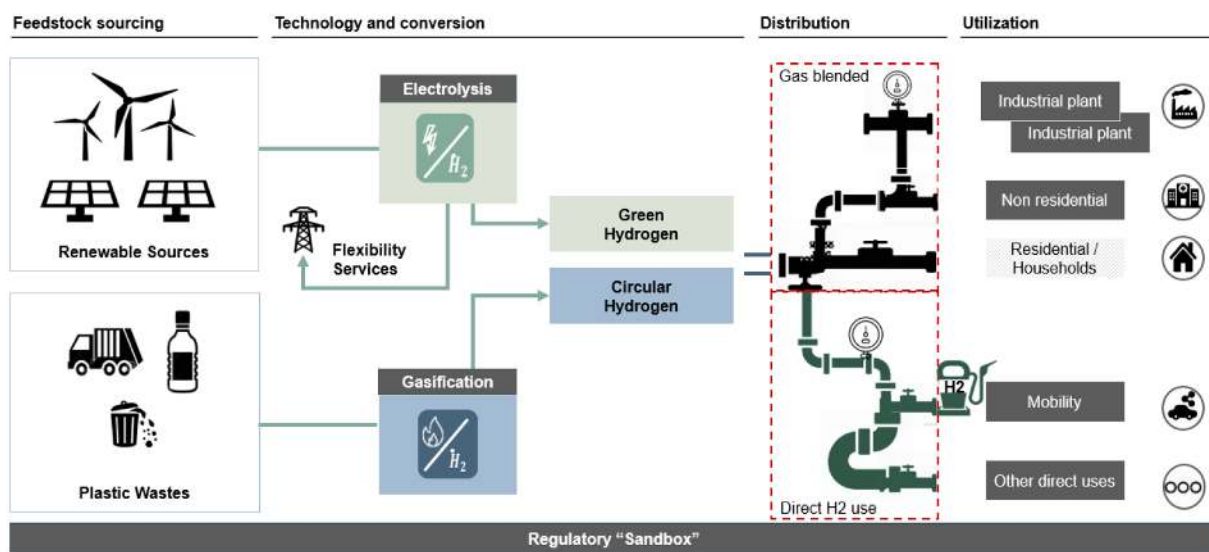
May 2019		
Geographical area	N° Plants	%
Northern Italy	170	63.7
Central Italy	60	22.5
Southern Italy and islands	37	13.9

May 2019		
Geographical area	kWel.	%
Northern Italy	41760	73.7
Central Italy	9552	16.9
Southern Italy and islands	5362	9.5

Italy - Initiative by NextChem & ENI - Circular District



<https://nextchem.it/nl/node/98>



NextChem ongoing studies to produce and therefore inject green H2 and syngas in the gas distribution networks.

➤ 10 June 2019

«Eni and Maire Tecnimont sign agreement to introduce new technology that transforms non-recyclable waste into hydrogen and methanol» (processes based on gasification are included)

➤ 23 July 2020

«NextChem and US carbon recycling company, LanzaTech sign an agreement to license the "Waste to Ethanol" process line» (syngas fermentation)

➤ January 2020

NextChem - «The role of Green Chemistry in accelerating and enabling the energy transition, via circularity, zero carbon and biobasedproducts» (NextChem is the Maire Tecnimont's company for green chemistry and energy transition technologie)

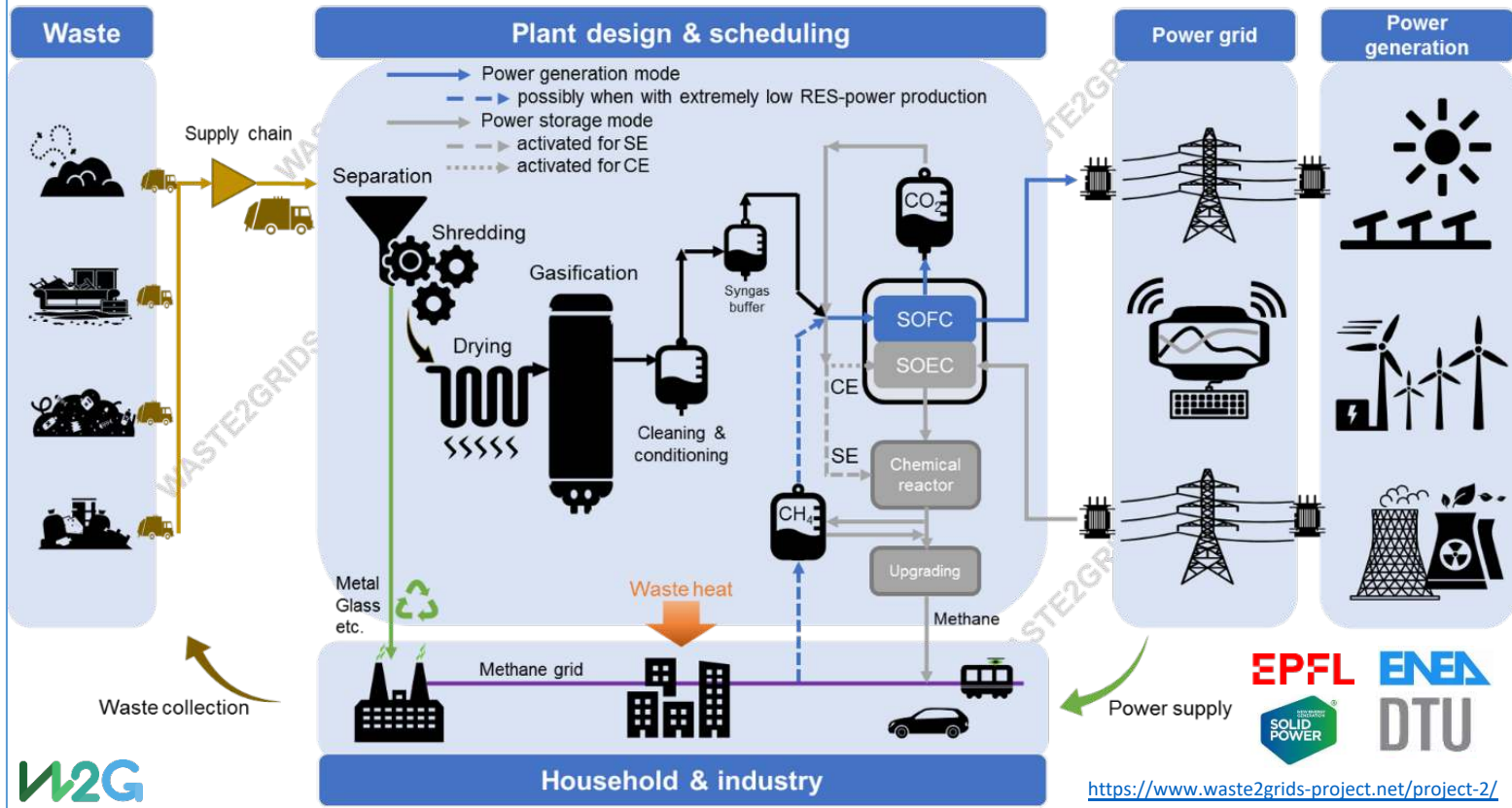
➤ By 2023

Planned to start-up a plant for «waste to methanol», 110 kTons/year capacity, in Livorno, Tuscany. (more info in the form uploaded in the Task database)

Italy - project Waste2GridS

A single, **dual-mode** plant for power storage and generation

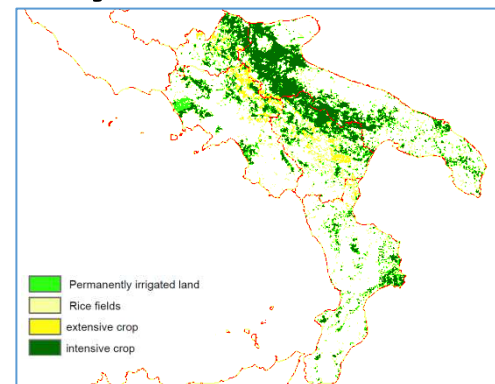
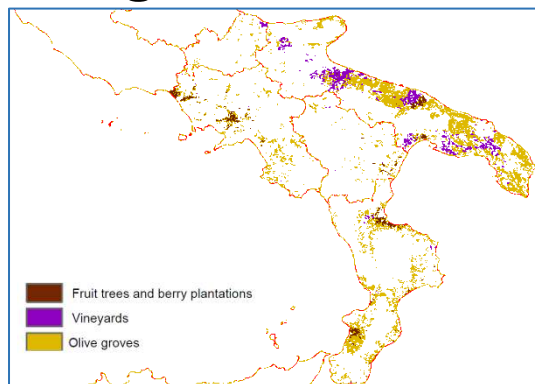
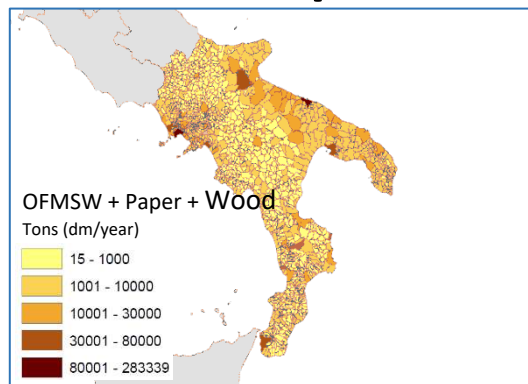
EU H2020 project WASTE2GRIDS: A contribution to circular economy



Converting WASTE to Offer Flexible GRID Balancing Services with Highly-integrated, Efficient Solid-oxide Plants

Italy - project Waste2GridS

The bio-waste potential of Agri – Residues – Italy – total 2030

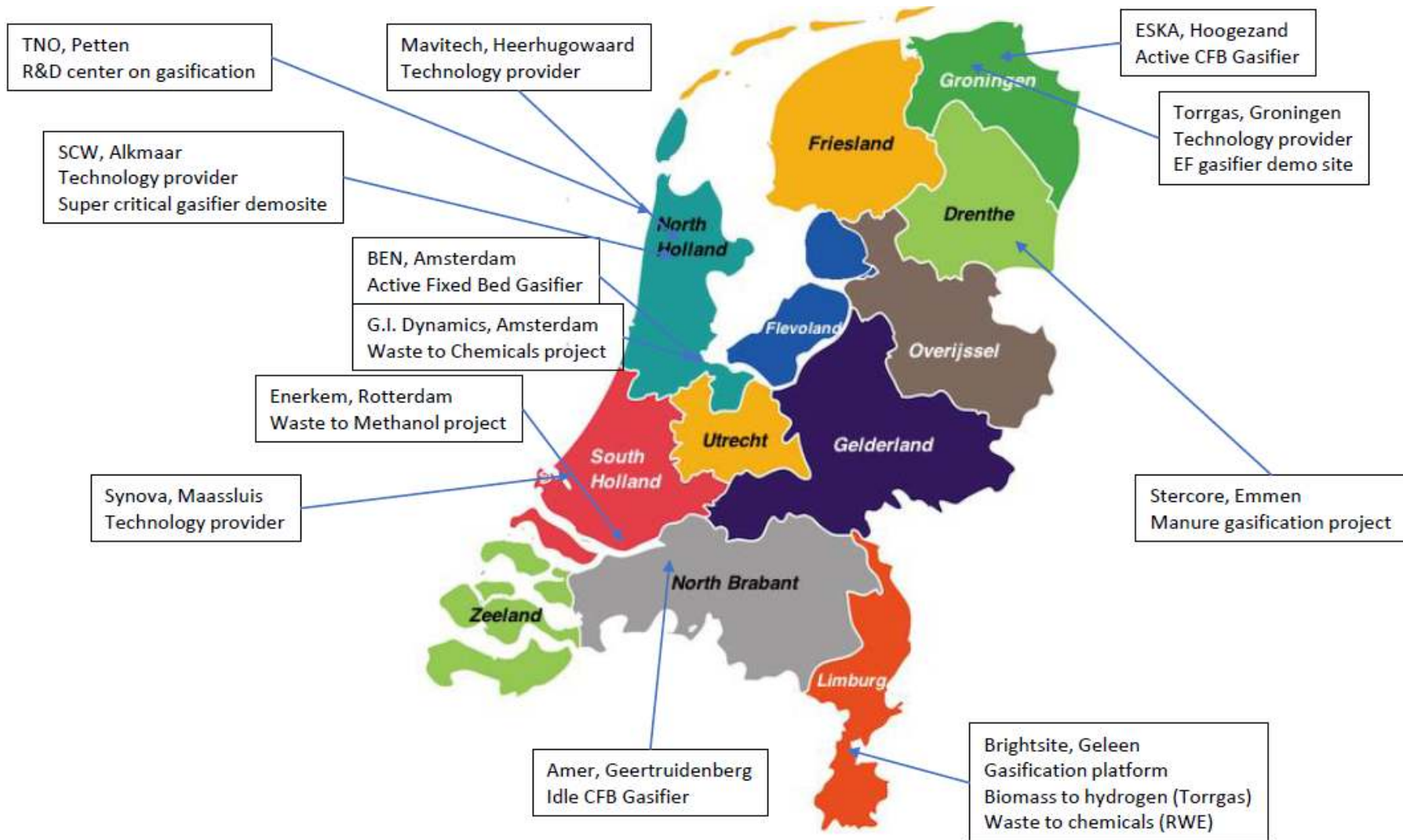


Feedstock		Availability in 2030 (t)	Moisture (%-wt)	Energy content	
				LHV (MJ/kg)	MWh (dry part)
Agricultural residues	Prunings	1,221,102		18.0	6,105,510
	Straw	826,384		17.2	3,948,279
MSW - separated at origin	Organic	3,721,023	50	20.7	10,697,941
	Paper	796,697	10	18.0	3,585,137
	Wood	86,110	10	18.0	387,495
	total	6,573,817			24,724,362

Design option ^{c)}	Biomass LHV input [kW]	Power Production Mode ^{a)}		Methane Production Mode ^{b)}			
		Electricity produced [kW]	Pow Gen eff [%]	Electricity consumed [kW]	Methane produced [kg/s]	Power stored in methane [kW]	Pow. Stor. eff [%]
1	50104	21862	43.6	24287	0.84	41959	56.4
2	50104	23756	47.4	48533	1.36	67836	68.8
3	50104	25149	50.2	47692	1.36	67836	69.4
4	50104	25564	51.0	19044	0.77	38588	55.8
5	50104	26986	53.9	49638	1.36	67879	68.1

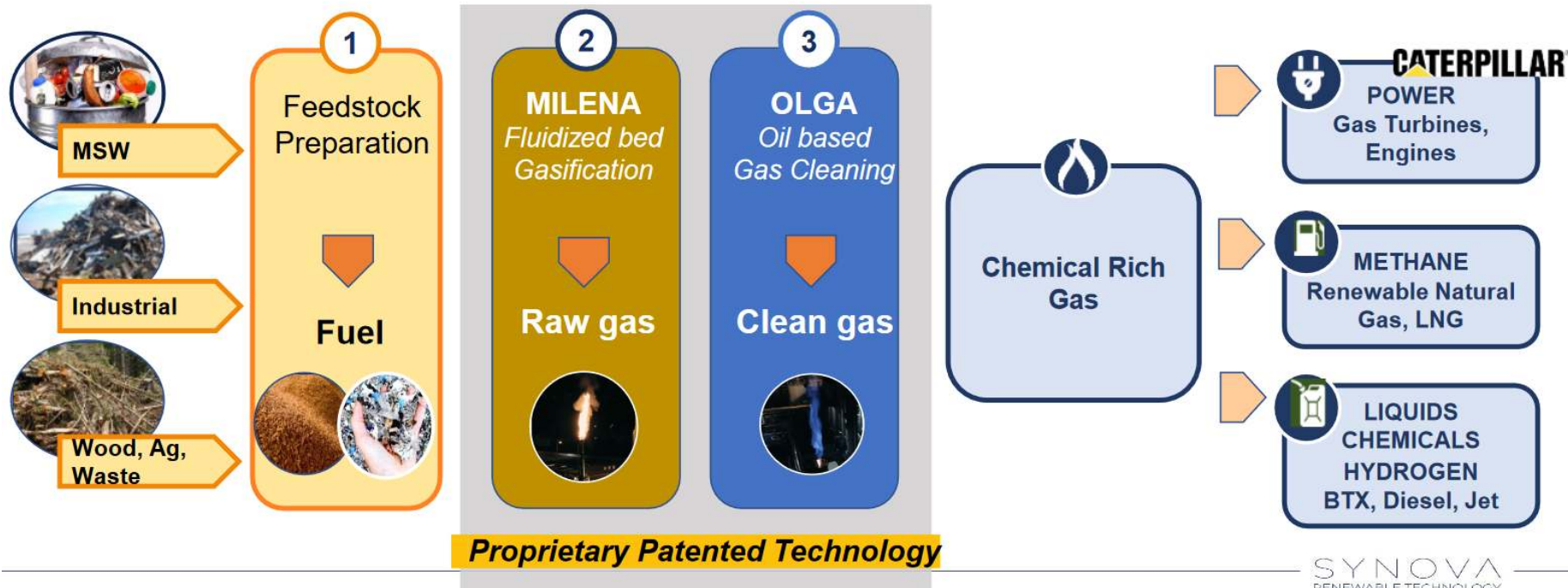
a) Via biomass gasification and use of the syngas in SOFC; b) the produced syngas is converted in CH₄ by addition of H₂ from electrolyzer (SOEC). The SOEC is operated using power excess from discontinuous RES.
c) Process chains designed according to the mode of plant operation coupled with methane production (power-to-methane (PowSto) mode integrated with steam electrolysis).

The Netherlands



The Netherlands

SYNOVA
RENEWABLE TECHNOLOGY



30 -60 kW_{th} lab unit including OLGA extensively tested

- 1 MW_{th} pilot/demo unit including OLGA extensively tested
- 4 MW_{th} demo/commercial unit including OLGA , tested

The Netherlands

Waste streams as feedstock



Scrap wood and wood processing residues



Grass, straw and agricultural residues

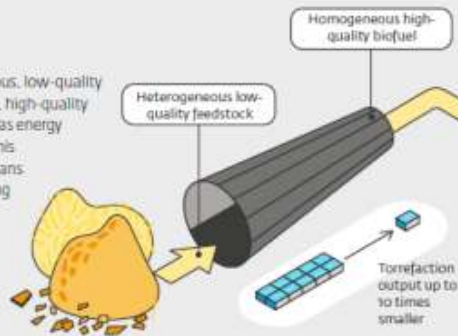


Wet waste streams such as manure and sewage sludge for wet torrefaction

Torrefaction processes use a wide range of waste streams that would otherwise be burned or left to perish. This greatly increases the amount of waste that can be reused.

Torrefaction

Torrefaction converts heterogeneous, low-quality waste streams into homogeneous, high-quality biofuels that are around ten times as energy dense as the original feedstocks. This enables efficient transport and means torrefaction is a vital link in enabling large-scale reuse of problematic waste streams.



Uses of green gas

The Torrgas process produces green gas from syngas. This gas is transported through gas infrastructure to users in the industrial domain (for use as a feedstock and for process heating) and to the built environment.



Industry & chemistry



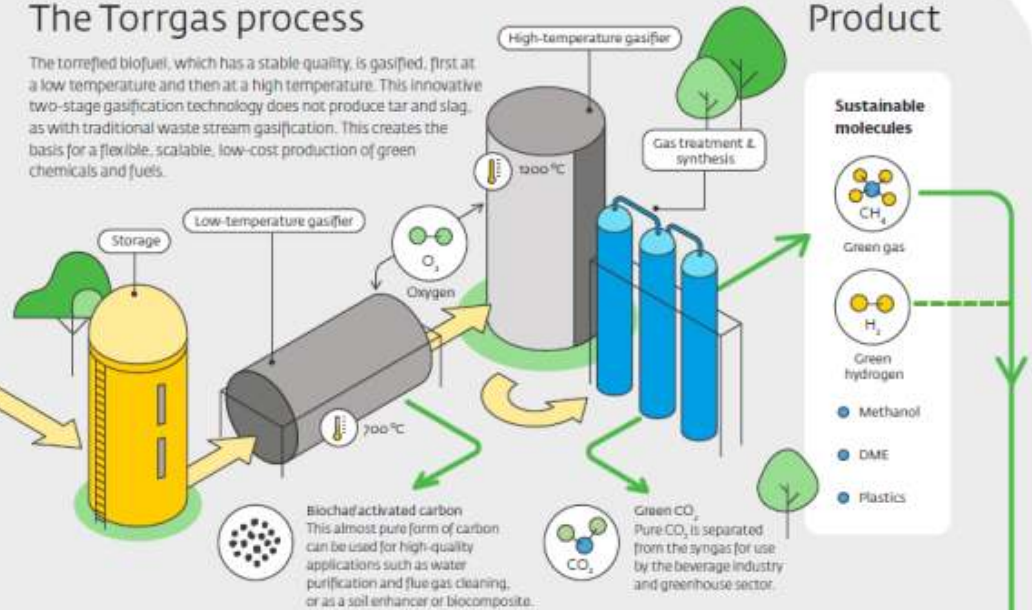
Built environment



Transport & mobility

The Torrgas process

The torrefied biofuel, which has a stable quality, is gasified, first at a low temperature and then at a high temperature. This innovative two-stage gasification technology does not produce tar and slag, as with traditional waste stream gasification. This creates the basis for a flexible, scalable, low-cost production of green chemicals and fuels.



Product

Sustainable molecules



Green gas



Green hydrogen

• Methanol

• DME

• Plastics

Benefits of the Torrgas process



Scalable

A Torrgas plant can be scaled up to 100 MW.



Affordable

Activities such as the scaling up and marketing of biochar and green CO₂ make it increasingly cheaper to produce syngas. So much so, in fact, that it can even compete with fossil alternatives on price.



Fully circular

Low-quality waste streams are fully converted into high-value molecules (syngas and green CO₂) and products (biochar).



CO₂ reduction

Waste streams are converted into usable products. This prevents combustion and carbon emissions, effectively removing CO₂ from the atmosphere.

torrgas | gasunie
closing carbon's energy

The Netherlands

RWE

Project to use waste streams for hydrogen production

Capacity will be 40 kton per year of hydrogen (1000 tpd of waste)

The whole technology chain will be built:

- Waste separation and drying
- Torrefaction of waste to uniform pellets
- Siemens gasification technology (EF) to produce syngas
- Hydrogen production and CO₂ production

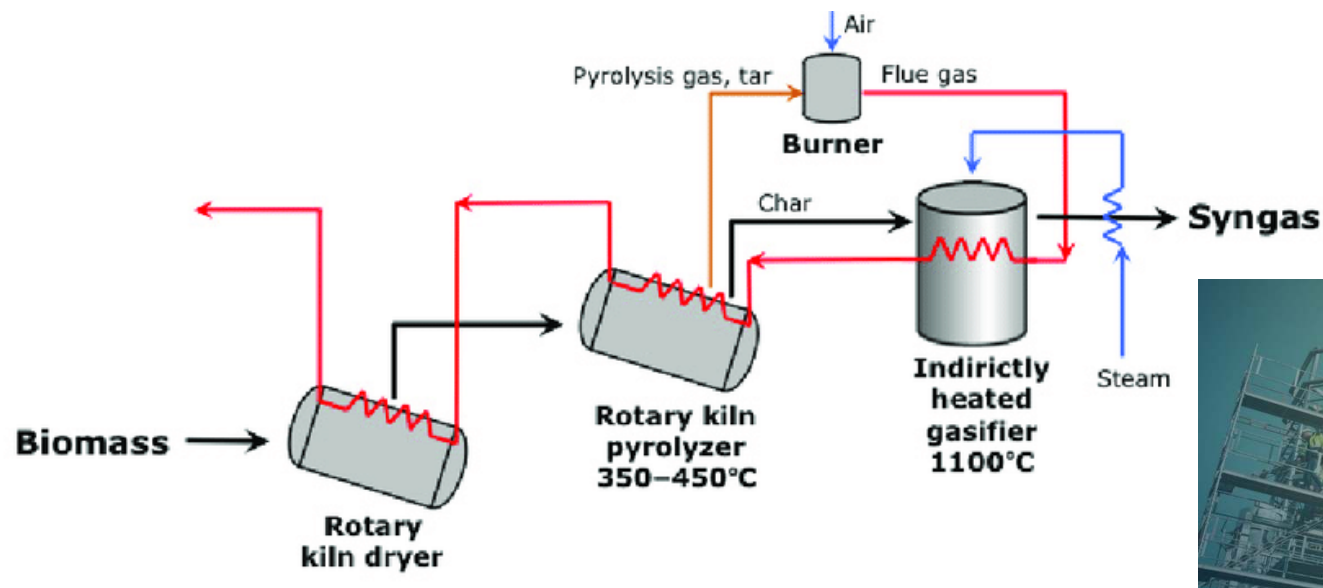
Location: Chemelot industrial site in Limburg

Process tested in Freiberg



Sweden

Cortus 6 MW WoodRoll® gasifier at the Höganäs steel plant



Cortus WoodRoll® in Mariposa (US)

Pacific Gas & Electric (PG&E) has approved Cortus and the Mariposa-project for delivery of 2,7 MW electricity.

Sweden

Cortus WoodRoll® & ENGIE (Fr)

Collaboration project (WoodHy) initiated 2019 on biomass-to-hydrogen project in Bordeaux, France

Cortus has a patented PSA configuration that results in about 15% more hydrogen output than other conventional systems

An order from **EngieCofely** worth 135 000 € to carry out an **Advanced Feasibility Study** of a plant for hydrogen and CO₂ production

More info at <http://cortus.se>



Sweden

MEVA CHP-plant in Hortlax

Entrained flow cyclone gasifier based on research at Luleå University of Technology



Output: 1.2 MW_{el} and 2.4 MW_{th}

Feedstock: small fraction fuels (sawdust, wood fibers and agricultural residues)

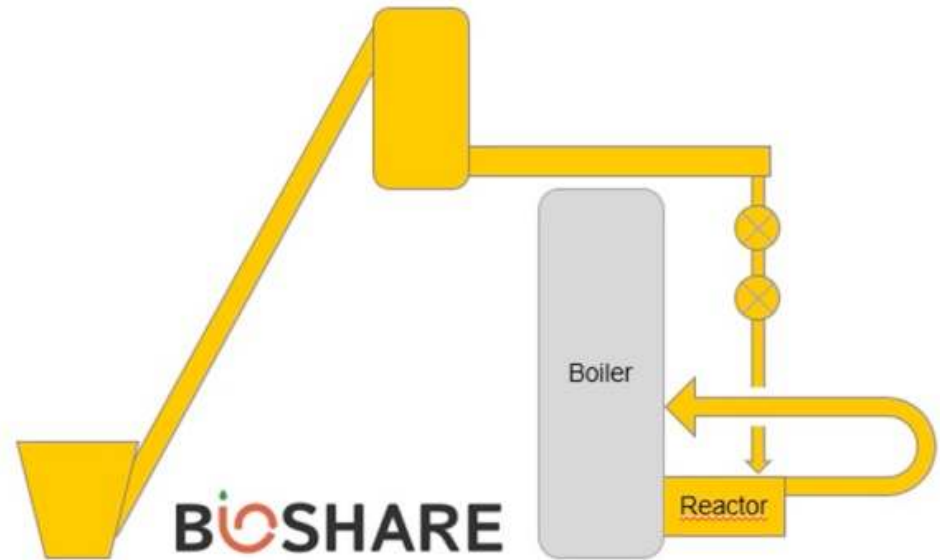
Applications: CHP, fossil process gas replacements and industrial drying processes

Joint project with Andritz-Enviroburners:
A complete gas burner test facility to enable engineering of industrial burner installations for process heat generation

More information at <http://mevaenergy.com>

Sweden

Project under development



The company BioShare has recently been granted funding ($\approx 1,7$ M€, 2020-2024) to demonstrate integrated gasification and pyrolysis at a CHP-plant

The reactor will be operated during three heating seasons to assess performance and gas quality with different feedstock

Also test different downstream units in pilot scale in separate projects with customers and other stakeholders.

Sweden

Projects on hold

The Gobigas plant in Gothenburg

Plant was in operation

Oct.2013 - May 2018

Now still moth-balled and for sale

Now in conservation state, but
will most likely be dismantled

LTU Green Fuels plant

Plant in conservation state

New regional funding secured for three years to keep the plant from
being dismantled and to find new projects

GoBiGas – step by step

- **Performance goals:**

- Biomass to biomethane 65 - 70%
- Energy efficiency > 90%

- **Phase 1:**

- Demonstration plant
- Evaluation, R&D programme
- 20 MW generating 160 GWh/year
- In operation early 2013
- Allothermal (in-direct) gasification

- **Phase 2:**

- 80-100 MW generating 640-800 GWh/year
- In operation after evaluation of Phase 1
- Technology not yet chosen

 Göteborg Energi



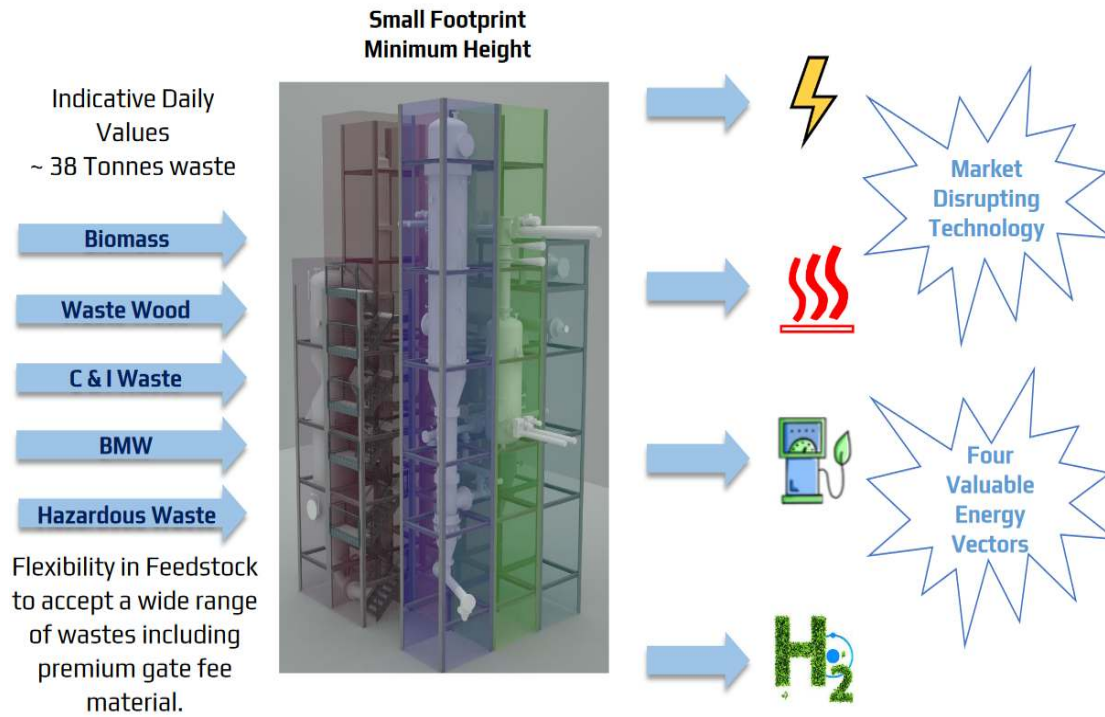
**Official start-up
October 28, 2013.**

United Kingdom

ETI/KEW demonstration plant (near Birmingham)



Waste gasification plant (1,5 MW, pressurised fluidised bed technology) started to produce clean syngas which can be used for the generation of electricity, heat, hydrogen and liquid bio fuels.



Status Nov. 2020:

Commissioning with
tests on waste wood,
RDF and other
feedstock finished.

United Kingdom

Advanced Biofuel Solutions Ltd (ABSL), Swindon site

Commercaill demo for production of 20-25 GWH/y of bio SNG

Status Nov. 2020

- Construction complete
- Cold commissioning finished
- Hot commissioning in January 2021
- Waste processing planed Feb./March 2021



<https://www.linkedin.com/company/advanced-biofuel-solutions-ltd/>

Project Bright

- Similar technologies to ABSL Swindon plant
- Larger scale 320 GWJ/y of bio SNG
- Located near centre of HyNet projects – CCS infrastructure should be available 2025

USA

Fulcrum Bioenergy/Sierra Biofuels Under construction



Gasification/FT under construction

- MSW feedstock/ TRI indirectly heated fluid bed steam reforming gasifier
- Input: 175 kt/y prepared MSW
- Johnson Matthey DAVY™/BP fixed-bed FT
- Start-up in Q1 2021
- Recently pivoted from jet fuel to FT wax
- Plant is 'full scale'; scale up (3X) planned with parallel trains
- 12 new projects currently planned/underway

USA

Red Rock Bio (Lakeview, Oregon) Under construction



Red Rock Biofuels Lakeview, OR: Aerial view on September 4, 2020, looking SW

- Pathway: gasification to Fischer-Tropsch
- Technology providers
 - Gasification: TCG Global steam reforming
 - FT technology: Velocys and Emerging Fuels Technology
- Feedstock: forest residue (136,000 tons per year)
- Products: cellulosic renewable jet, diesel, and naphtha fuels
- Offtakes: Southwest Airlines, FedEx Express
- Capacity: 15 MGY
- Anticipated construction completion: Q1 2021

France



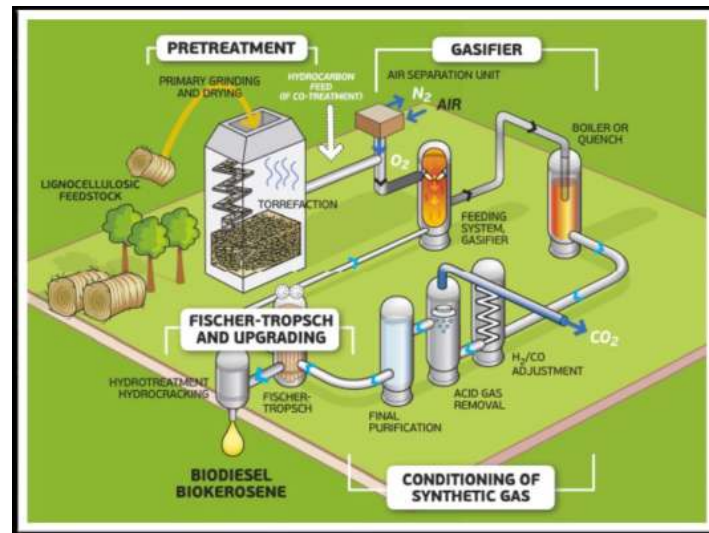
Objective: develop, demonstrate and commercialize a full B-XtL chain

- Process chain validated and optimized on a wide range of biomass
- End to end solution: From R&D to market / From biomass to final products



Status November 2020:

- ✓ Commissioning and start up completed
- ✓ Ongoing technology optimisation
- ✓ Commercialization: 1st contacts
 - Axens (on behalf of consortium): single licensor for the complete B-XtL chain
 - A performance guarantee for the complete chain:
 - For processes and catalysts
 - From biomass to final products
 - A full scope of services from studies to unit start-up and follow-up



<https://www.ifpenergiesnouvelles.com/innovation-and-industry/our-expertise/renewable-energies/biofuels/our-solutions>

<https://www.total.com/energy-expertise/projects/bioenergies/biotfuel-converting-plant-wastes-into-fuel>

France



Objective: Demonstrate the feasibility of the production of 2nd generation bioCH₄

- A 10-year R&D program
- Demonstration platform covering the entire production chain

Feedstock:

- woody biomass
- non-recyclable waste (Solid Recovered Fuel)

Output:

- biomethane 0,4 MW (demo)

Status december 2020:

- ✓ Oct. 2017: platform inauguration
- ✓ Nov. 2018: 1st production of purified synthesis gas from biomass
- ✓ Nov. 2019: 1st production of biomethane from woody biomass
- ✓ Nov. 2020: 1st production of biomethane from SRF



France / Belgium

NOTAR[®] Reactor

A MULTI-STAGE DOWNDRAFT GASIFIER

ADVANTAGES

- ✓ CLEAN GAS
(99,95% TARS DESTROYED)
- ✓ COMPACT
GASIFICATION UNIT
- ✓ SIMPLE AND
COST EFFECTIVE GAS
CONDITIONING UNIT

SYNGAS ✓ NO TAR
NO HEAVY METAL
NO POLLUTANT

TECHNOLOGICAL ADVANTAGE

Accurate control of operating parameters:

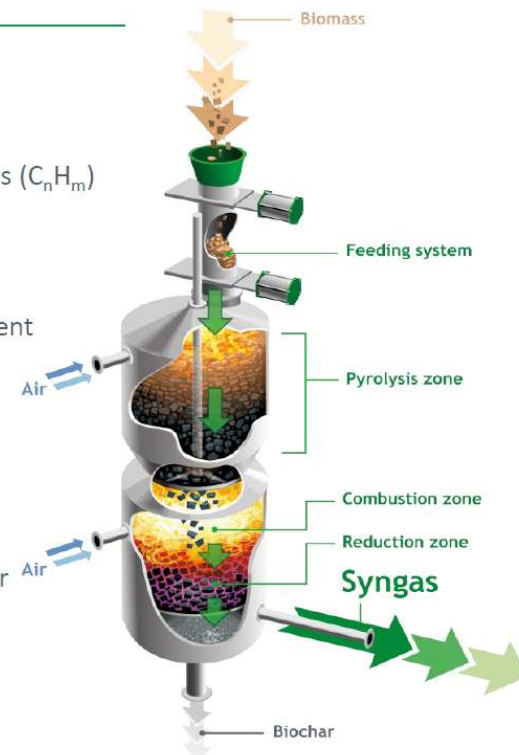
- ✓ Tar Free Char (C)
- ✓ Light Pyrolysis Gases (CH_p)
- ✓ Primary Tar concentrated in pyrolysis gases (C_nH_m)

Combustion in gaseous phase:

- ✓ Destruction of Pyrolysis Tar
- ✓ Conversion of biomass with high ash content
- ✓ Power range from 0.1 to 2 MW

Reduction is fed with Tar Free products:

- ✓ Production of a Tar Free syngas
- ✓ Maximum Gas Temperature of 750°C
- ✓ Metallic Compounds condensed in biochar



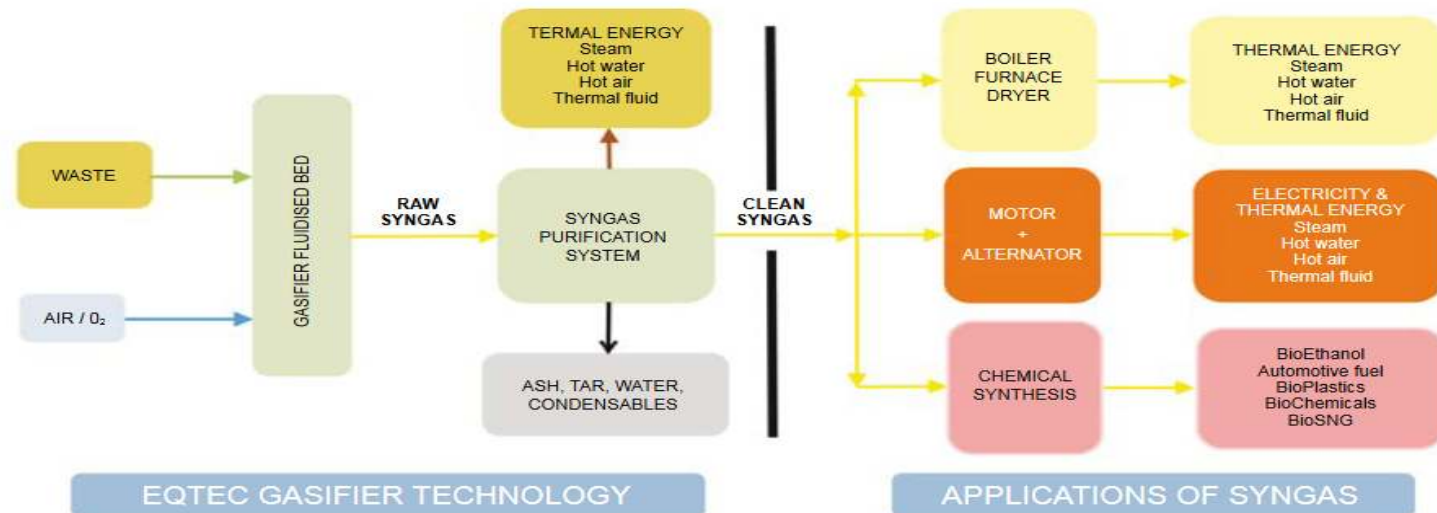
Biochar as a valuable by-product



	Unit	Dry base
Humidity	%	0
Mineral materials rate	%	15 – 20
Volatile matter rate	%	< 5
Fixed carbon rate	%	75 – 80
LHV	kJ/kg	25 000 – 28 000



Spain



EQTEC



Location	Cuidad Real, Spain
Description	5,9 MWe IBGPP
Electrical efficiency	28%
Total efficiency (electrical + thermal)	64%
Feedstock	Olive Mill Pomace
Feedstock throughput	4 ton/h
Engine	3 x GE Jenbacher 620
Operating temperature range	720-750 °C
Operating hours	111,000+ (third party certificates available)

Spain

Bubbling fluidized bed technology

Feedstock: wood chips, meat and bones meal



Reactor	Bubbling Fluidised Bed
Thermal power	7,4 MWth (2 MWe)
Application	CHP Electricity production 1,6 Mwe (GE Jenbacher JMS – 320 (x3)) Heat used for drying in briquettes manufacturing (variable heat required)
Gasifying agent	Air
Operation Pressure / Temperature	0,3 barg / 800°C
Biomass load	45 t/day
Start up date	2012
Biomass	Wood chips
Location	Ejea de los Caballeros (Zaragoza)

Reactor	Bubbling Fluidised Bed
Thermal Heat	10 MWth
Use	12 T/h saturated steam 10 bar
Gasifying agent	Air
Operation Pressure / Temperature	0,3 barg / 800°C
Start up date	2013
Biomass	Meat and bone meal (MBM) Cat01
Location	Aldeaseca de la Frontera (Salamanca)

Reactor	Bubbling Fluidised Bed
Thermal Heat	16 MWth
Use	20 T/h saturated steam 10 bar
Gasifying agent	Air
Operation Pressure / Temperature	0,3 barg / 800°C
Start up date	2014
Biomass	Meat and bone meal (MBM)
Location	Sao João da Madeira (Porto, Portugal)



Spain



Plant in Santa Perpetua de la Mogoda, Barcelona	
Application	Steam generation for industrial processes
Feedstock	Forest biomass, plastics, RDF, waste biomass
Output	25 t/h of steam
Start up	2018
Output/Capacity	20 MWth



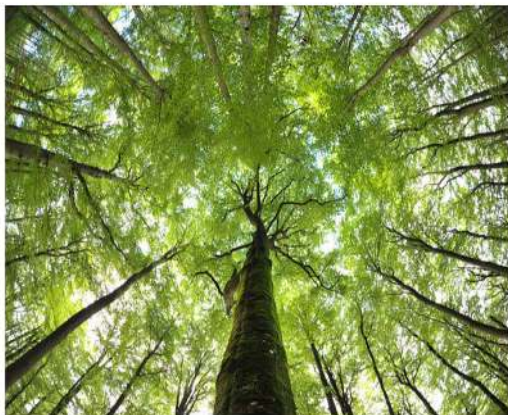
Plant in Villacañas, Toledo	
Application	Power generation through steam turbine 8 MWeI
Feedstock	Forest biomass
Technology	Wood gasification, syngas combustion
Start up	2010
Steam boiler output	44 tons/h 460°C, 40 bar

IEA Bioenergy Task 33

Status reports

Status report on thermal biomass
gasification in countries participating
in IEA Bioenergy Task 33

2016



IEA Bioenergy Task 33

Special report on emerging gasification technologies



Emerging Gasification Technologies for Waste & Biomass

IEA Bioenergy: Task 33
December 2020



Photo by courtesy of Meria Energy AB

Technology Collaboration Programme
by IEA

Will be published on Friday 26. February 2021

At the Task 33 website in section „Task projects“

Conclusions

Thermal gasification makes sense!

- Variety of feedstock can be utilized and waste disposed
- Different gasification technologies available in commercial scale
- Variety of possible products
 - Syngas for CHP
 - Biofuels
 - Biochemicals
 - Biochar (CCS) - negative CO₂ footprint
 - ...
- Integration/combination of gasification with other technologies

Conclusions

What is needed now?

To recognize that:

- gasification based on biomass and waste can be sustainable technology saving our environment.
- the combination of gasification with other technologies makes it much more attractive.

Clear and stable political frame, supporting renewable energy and products to encourage investors and implementation of know how on gasification!

Thank you!

Jitka.hrbek@boku.ac.at



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Technology Collaboration Programme

www.ieabioenergy.com